Task 4 - Weather Logger

Project Title

Weather Logger using Sensors and Arduino

Objective

To design a system that reads environmental data from sensors and stores the data for monitoring and analysis.

Duration

6 Hours

Project Complexity

Easy

Learning Outcomes

- Learn environmental sensor interfacing.
- Understand data logging concepts.
- Monitor temperature and humidity in real time.
- Store sensor data for later analysis.

Components Required

- Arduino Uno
- DHT11 / DHT22 Sensor
- BMP280 (optional pressure sensor)
- SD Card Module
- SD Card
- Jumper Wires
- Breadboard
- LCD Display (optional)
- Power Supply / USB Cable

Working Principle

1. Sensors measure temperature and humidity.
2. Arduino reads data at fixed intervals.
3. Data is written to SD card with timestamp.

4. Stored logs can be analyzed later.
5. Process repeats continuously.

Circuit Connections

- DHT Data -> D2
- DHT VCC -> 5V
- DHT GND -> GND
- SD Module CS -> D10
- SD Module MOSI -> D11
- SD Module MISO -> D12
- SD Module SCK -> D13
- SD VCC -> 5V
- SD GND -> GND

Arduino Code

```
#include <DHT.h>
#include <SPI.h>
#include <SD.h>

#define DHTPIN 2
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
File dataFile;

void setup() {
  Serial.begin(9600);
  dht.begin();

  if (!SD.begin(10)) {
    Serial.println("SD Card Failed");
    return;
  }
  Serial.println("SD Card Ready");
}

void loop() {
  float temp = dht.readTemperature();
  float hum = dht.readHumidity();

  dataFile = SD.open("weather.txt", FILE_WRITE);

  if (dataFile) {
    dataFile.print("Temp: ");
    dataFile.print(temp);
```

```
    dataFile.print(" C, Humidity: ");
    dataFile.print(hum);
    dataFile.println(" %");
    dataFile.close();
  }

  Serial.print("Temp: ");
  Serial.print(temp);
  Serial.print(" C  Humidity: ");
  Serial.print(hum);
  Serial.println(" %");

  delay(5000);
}
```

Sample Output

```
Temp: 29 C  Humidity: 64 %
Temp: 30 C  Humidity: 62 %
```

Real-World Applications

- Personal weather stations
- Agricultural climate monitoring
- Greenhouse monitoring
- Research data collection

Future Enhancements

- RTC timestamp module
- Cloud IoT dashboard
- Rainfall and wind sensors
- Mobile alerts

Conclusion

This project collects and stores weather data automatically. It is useful for monitoring environmental conditions and research purposes.